

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458930.8399 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.068 +/- 0.042
Transit depth (Rp/Rs)^2	0.46 +/- 0.58 [%]
Orbital Inclination (inc)	86.56 +/- 2.26
Airmass coefficient 1 (a1)	0.26 +/- 0.17
Airmass coefficient 2 (a2)	1.17 +/- 0.94
Scatter in the residuals of the lightcurve fit is	67.15 %
Best Comparison Star	#1 - [574, 225]
Optimal Aperture	2.88
Optimal Annulus	5.75
Transit Duration (day)	0.065 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.068 ± 0.042 vs 0.1063 (deviation 0.78σ)
Duration fit vs published	0.065 d vs 0.0867 d (deviation 0.6σ)
Mid-time O–C	8.7 min
Depth SNR	1.4
Residual scatter	67.15 %

Light curve

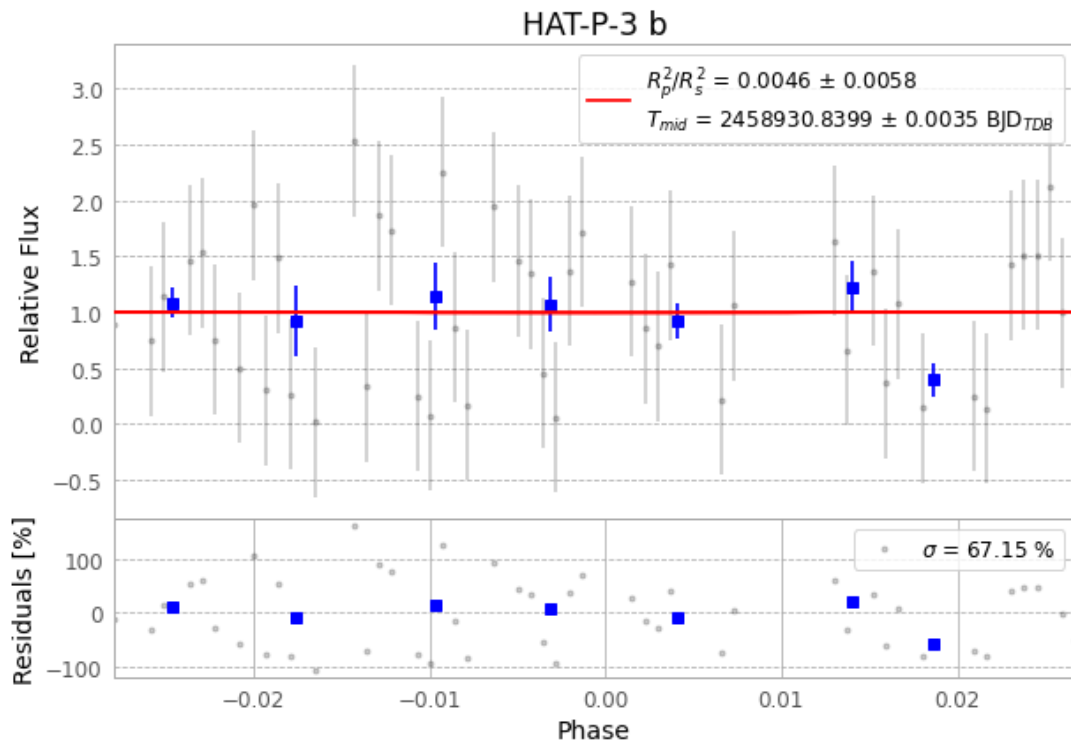


Figure 1 — FinalLightCurve_HAT-P-3 b_2020-03-21.png (SHA-256 3ad981b917825455...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [609, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e6b54f1a46f21827...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), HATP-3200322060612.FITS → HATP-3200322095712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-200322.log (79b5c9102320...), FinalLightCurve_HAT-P-3 b_2020-03-21.png (3ad981b91782...), FinalLightCurve_HAT-P-3 b_2020-03-21.csv (f1644fb34905...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:33Z.